

Name: _____ **Date:** _____

Problem 1. Find any inflection points of the following functions (hint: make a sign chart of the second derivative).

a) $f(x) = x^3$

b) $f(x) = x^3 - 9x^2 + 7x$

c) $f(x) = 2x^3 + 3x^2 - 12x + 1$

d) $f(x) = x^4 + 4x^3 + 1$

e) $f(x) = \sin(x)$

f) $f(x) = \cos^2(x)$

Problem 2 (Optional, challenge exercise). Let $f(x) = e^{-x^3+3x}$.

a) Find the second derivative $f''(x)$

b) Find all values of x where $f''(x) = 0$